For both models, the forecast error variance decomposition of y indicates that shocks to y itself dominate both in the short and in the long-run. According to the VAR(1), the effect grows in time from 83% to 85% of the forecast error variance, while with the VAR(4) the effect is more pronounced in the short term, 84%, and it decreases to 80% after 10 periods. The shocks to x contribute much less in explaining y, and they behave in a specular way with respect to the y shocks. Finally, the forecast error decomposition of x in both figures shows that also for x its own shock dominates in the short run, but the effects of the y shock increase over time while those of x decrease, so that the two shocks have comparable importance at longer horizons.

6.10 Empirical examples

6.10.1 GDP growth in the Euro area

In this example we apply the VAR estimation and forecasting techniques we saw in the theoretical sections of the chapter to study the relationship among the GDP growth rates of three major countries in the Euro area, namely, France, Germany and Italy, using the United States as an exogenous explanatory variable. The data set contains 108 quarterly observations from 1986Q1 to 2012Q4. The variables are the growth rates of real GDP expressed as percentage change from the previous year (%yoy), downloadable from DATASTREAM.[1] The series are all adjusted for seasonality and labeled as g_i where $i = fr, ger, it, us$; the dataset also contains some binary dummy variables labeled D_19XXqx accounting for exceptional episodes in the chosen temporal period.

As we can see from Figure 6.10.1, the three growth rates seem to be tightly linked, suggesting a synchronized business cycle through the entire Euro area, or at least its major economies. The gray shaded areas indicate the recessions occurring in the United States: the one that occurred in 1991 impacted our Euro area countries only in 1992 - 3, after the break-up of the European Monetary System (EMS), while the 2007 financial crisis arrived in Europe with only a few quarters delay, qualifying indeed as a "global" crisis.

[1]Note that the GDP growth rates can be manually calculated by using the log difference operator on the series of real GDP in levels. In this example we chose to use the (%yoy) growth rates calculated by Oxford Economics just because they are available from the 1980s on, while the real GDP series in levels are available only from the 1990s.

From the graph it also emerges that for the two main recessions of 1992 and 2008, Germany and Italy display more pronounced troughs than France, and this can be due to the tight commercial links existing between the German and Italian economies and their degree of openness.

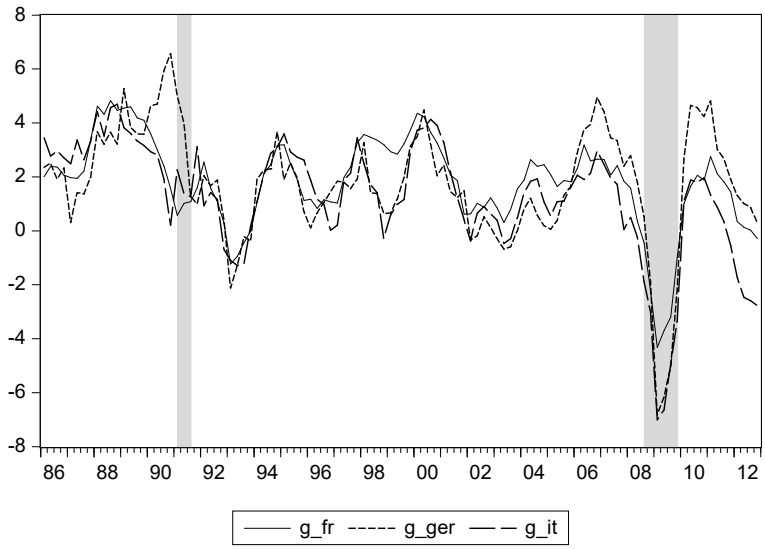

Figure 6.10.1: *Real GDP growth rates for France, Germany, and Italy during the period 1986Q1 - 2012Q4. The gray shaded areas indicate the NBER recessions for the United States*

Preliminary analysis on the series confirmed that g_fr, g_us, and g_ger can be considered as stationary at the 5% confidence level, while g_it cannot, and it is probably characterized by a unit root. However, from an economic point of view, it is surprising that the growth rate is integrated, and therefore we will treat it anyway as stationary.

The next step of our analysis is to see whether there are common intertemporal dependencies among the growth rates of the three Euro area countries, using g_us as an exogenous variable, due to the global importance of the United States and the limited effects of each of the Euro area countries on its large economy. First, we will specify suitable VAR models and estimate them over the period 1986 - 2003, next we will evaluate their forecasting performance from 2004 to 2006. Then, we will repeat the estimation until 2006Q4 and evaluate the forecasting performance of the model during 2007 - 2012, including the crisis period.

VAR Lag Order Selection Criteria

Lag	*LogL*	*LR*	*FPE*	*AIC*	*SC*	*HQ*
0	-324.200	NA	3.034	9.623	9.721	9.662
1	-217.244	201.331	0.170	6.742	7.134*	6.898
2	-198.730	33.217*	0.129*	6.463*	7.148	6.734*
3	-193.902	8.233	0.146	6.585	7.564	6.973
4	-183.820	16.310	0.143	6.553	7.827	7.058

* Indicates lag order selected by the criterion. LR: sequential test (each at 5% level), FPE: final prediction error, AIC: Akaike information criterion, SC: Schwarz, HQ: Hannan-Quinn

Table 6.10.1: *Choice of the appropriate lag length p through information criteria*

Let us start by selecting the appropriate lag length p for our VAR model. Table 6.10.1 reports the information criteria associated with the various lags of our VAR model. The AIC, the Hannan-Quinn criterion, the Final Prediction Error (FPE), and the LR criteria all suggest $p = 2$. while the more parsimonious BIC indicates $p = 1$. As a consequence, we will estimate two rival VAR models with, respectively, $p = 1$ and $p = 2$, and we will evaluate their relative performance.

Note that both models include the same set of exogenous variables, namely, the growth rate of US GDP and the dummy variables necessary to remove some outliers (in 1988Q1, 1993Q1, and 2000Q1). Table 6.10.2 reports the estimation results of the VAR(1) model. VAR(2) results are not reported. Both VARs display a good overall performance given the not so high number of observations in the sample, and in general the exogenous variables we chose are significant. On the other hand, while in the VAR(1) model the significance of some lagged variables is very low, the same is generally not true for the VAR(2) model, suggesting that the former is probably misspecified. The same impression emerges when we look at the residuals of the two models. Tables 6.10.3 - 6.10.5 report results from the diagnostic tests for the VAR(1); results for the VAR(2) are not reported since they are very similar. The residuals of both models look quite "clean," although from their correlograms (not shown here), it is even more evident that the VAR(2) captures some autocorrelation that the VAR(1) does not. On the other hand, there is no sign of heteroskedasticty or non-normality in any of

the two models' residuals, and this is partly due to the dummies we have introduced. Finally, note that the inverse of the AR roots are all well within the unit circle, supporting our choice of treating the growth rate of Italy as stationary.

How are the variables dynamically related? We can answer this question through the Granger causality test, and to perform it we chose to use $p = 2$ lags.

From Table 6.10.6, it seems that the shocks are not significantly dynamically transmitted from Germany to France and Italy, while the reverse is not true and there are cross-linkages across France and Italy.

As the Granger causality test only considers lagged relationships, in order to better understand if there is correlation among contemporaneous shocks in our VAR(2) model, and hence contemporaneous correlation across the variables after conditioning on their lags, we can use an indirect method. In the absence of cross equation correlations, the ordering of the variables should not affect very much the behavior of the impulse response functions obtained with a Cholesky factorization of the variance covariance matrix of the VAR residuals. Hence, we can check whether this is the case.

We have tried two different orderings of variables to create the impulse response functions from the estimated VAR(2). The responses are rather similar for the two cases, here we present only results from the first order in Figure 6.10.2. Results suggest that indeed the contemporaneous correlation is limited, in line with economic reasoning that suggests that it takes some time for the shocks to be transmitted across countries. The response functions also indicate that there are positive spillovers across all countries, some of which become significant only 2 - 3 quarters after the shock.

Which of the two estimated models (VAR(1) or VAR(2)) will perform better from a forecasting point of view? We answer the question first for the forecast sample, 2004 - 2006. Then, we re-estimate the models until 2006 and use the second forecast sample, spanning from 2007 to 2012, to assess the robustness of the results and the performance of the models during problematic times.

The stochastic one-step ahead forecasts for the period immediately before the crisis are shown in Figure 6.10.3. The predicted series capture the overall behavior of the actual variables, though not their smaller shifts, and this is a common characteristic of both the deterministic and the stochastic forecasting methods. The detailed forecast metrics confirm the graphical impression that the VAR(2) is a little bit more precise in the prediction of the German

Vector Autoregression Estimates

	G_(FR)	G_(GER)	G_(IT)
G_FR(-1)	0.668	0.007	0.063
	(0.074)	(0.158)	(0.135)
	[9.064]	[0.047]	[0.470]
G_GER(-1)	0.009	0.745	0.122
	(0.042)	(0.089)	(0.076)
	[0.214]	[8.352]	[1.606]
G_IT(-1)	0.107	0.091	0.623
	(0.057)	(0.121)	(0.104)
	[1.893]	[0.751]	[6.008]
G_US	0.172	0.105	0.099
	(0.030)	(0.065)	(0.055)
	[5.695]	[1.632]	[1.794]
D_1988Q1	1.068	0.838	1.213
	(0.301)	(0.644)	(0.550)
	[3.551]	[1.301]	[2.206]
D_1993Q1	-1.624	-2.141	-1.604
	(0.310)	(0.663)	(0.566)
	[-5.247]	[-3.229]	[-2.834]
D_2000Q1	0.711	0.723	1.079
	(0.304)	(0.651)	(0.556)
	[2.339]	[1.110]	[1.942]
Standard errors in () and t-statistics in []			
R-squared	0.924	0.761	0.776
Adj. R-squared	0.917	0.739	0.755
Sum sq. resids	10.809	49.584	36.139
S.E.equation	0.411	0.880	0.751
F-statistic	129.747	33.982	37.044
Log likelihood	-33.923	-88.000	-76.771
Akaike AIC	1.153	2.676	2.360
Schwarz BIC	1.376	2.899	2.583
Mean dep	2.202	1.923	1.898
S.D.dependent	1.426	1.722	1.520
Determinant resid covariance (dof adj.)			0.066
Determinant resid covariance			0.048
Log likelihood			-194.456
Akaike information criterion			6.069
Schwarz criterion			6.738

Table 6.10.2: *Estimation results of the VAR(1) for the period 1986 - 2003*

Lags	LM-Stat	Prob
1	19.908	0.018
2	12.179	0.203
3	24.141	0.004
4	31.144	0.000
5	14.147	0.117
6	12.920	0.166
7	17.845	0.037
8	11.977	0.215
9	13.564	0.139
10	16.843	0.051
11	10.129	0.340
12	11.498	0.243

Table 6.10.3: *VAR(1) residuals serial correlation LM tests*

VAR Heteroskedasticity Tests:
Joint test:

Chi-sq	df	Prob.
72.340	66	0.277

Individual components:

Dependent	R-squared	F(11,59)	Prob.	Chi-sq(11)	Prob.
res1*res1	0.113	0.684	0.748	8.031	0.711
res2*res2	0.174	1.132	0.354	12.373	0.336
res3*res3	0.132	0.812	0.628	9.339	0.591
res2*res1	0.203	1.365	0.214	14.406	0.211
res3*res1	0.130	0.804	0.636	9.252	0.599
res3*res2	0.300	2.302	0.020	21.321	0.030

Table 6.10.4: *VAR(1) residuals heteroskedasticity test*

and French growth rates, but not of the Italian one.

As we expected, re-estimating the two VAR models in the longer sample 1986 - 2006 and producing forecasts for the period 2007 - 2012 proves more challenging. We present the forecasts in Figure 6.10.4.

For all the countries, both the RMSFE and the MAFE are much larger

VAR Residual normality Tests

Component	Skewness	Chi-sq	df	Prob.
1	-0.040	0.019	1	0.890
2	0.144	0.246	1	0.620
3	-0.333	1.311	1	0.252
Joint		1.576	3.000	0.665

Component	Kurtosis	Chi-sq	df	Prob.
1	2.541	0.623	1	0.430
2	3.543	0.873	1	0.350
3	4.112	3.661	1	0.056
Joint		5.157	3.000	0.161

Component	Jarque-Bera	df	Prob.
1	0.642	2	0.725
2	1.118	2	0.572
3	4.972	2	0.083
Joint	6.732	6	0.346

Table 6.10.5: *VAR(1) Residuals normality test*

Pairwise Granger Causality Tests

Null Hypothesis:	Obs	F-Statistic	Prob.
G_GER does not Granger cause G_FR	70	0.605	0.549
G_FR does not Granger cause G_GER		4.449	0.016
G_IT does not Granger cause G_FR	70	6.403	0.003
G_FR does not Granger cause G_IT		6.275	0.003
G_IT does not Granger cause G_GER	70	6.511	0.003
G_GER does not Granger cause G_IT		0.780	0.463

Table 6.10.6: *Granger causality test among the three variables using the lag length* $p = 2$

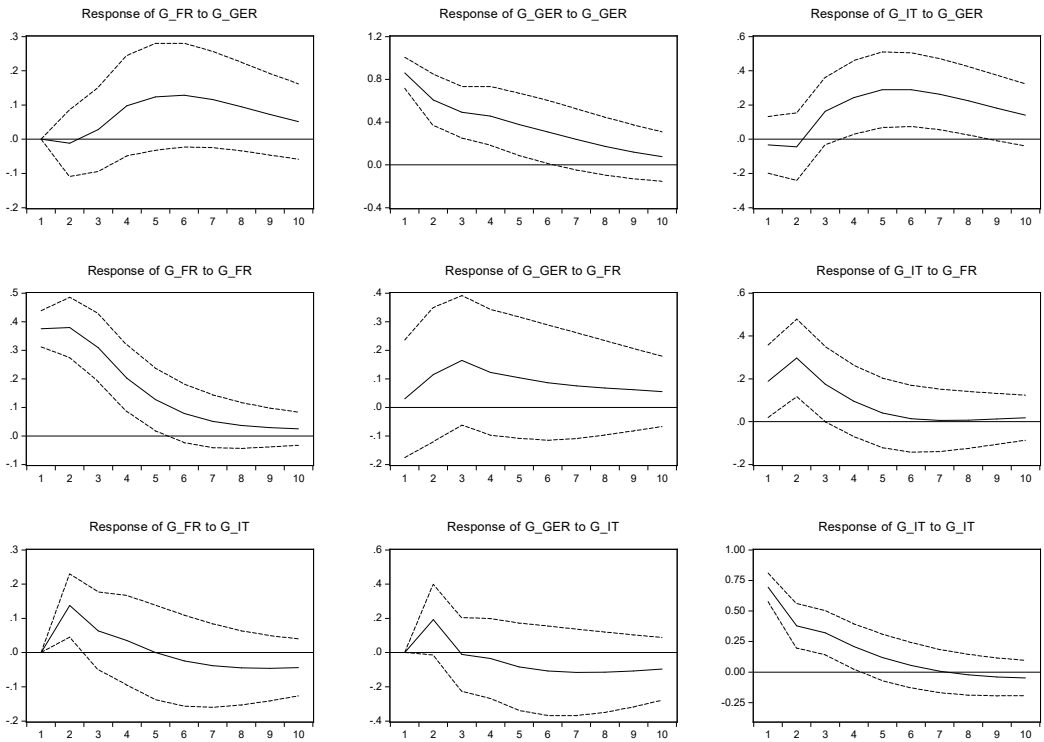

Figure 6.10.2: *Impulse response function from the VAR(1) model with the ordering (g_fr,g_ger,g_it)*

than during the first evaluation sample, often more than twice as large. Capturing the sharp decrease in the growth rates of the three countries' GDP growth is indeed quite complex, and even more sophisticated models could not achieve this goal at the time of the crisis. However, also in this sample the VAR(2) model yields more accurate predictions than the VAR(1) for all countries but Italy.

6.10.2 Monetary transmission mechanism

We can now analyze through a VAR model the transmission mechanism of monetary policy, describing how policy-induced changes in the nominal money stock or the short-term nominal interest rate impact real variables. In this example, we use US monthly data from 1986 until 2012.The dataset